

✅ Block-by-Block Explanation (Simple)

Block 1

Imports all required Python libraries (NumPy, Pandas, Matplotlib, Streamlit, etc.).
Also sets Streamlit page title.

“We load all tools we will use: data processing, GitHub API requests, embeddings, clustering, and UI.”

Block 2

Streamlit settings repeated (harmless duplicate).
Sets title of app.

Block 3

Gets GitHub token from Streamlit secrets.

“We read a GitHub token so we can download pull-request data.”

Block 4

Helper to call GitHub API safely + handle rate limits.

“This function sends requests to GitHub and waits if we hit rate limits.”

Block 5

Fetches list of merged PRs from repo.

“Collects PR metadata like PR number, status, dates.”

Block 6

Converts each PR into one big text string.

“We build one combined text block per PR: title + body + commit messages + file names.”

This text will later be embedded.

Block 7

Fetch full details for each PR — files, commit messages, labels, etc.

“Downloads deeper PR info, not just summary.”

Block 8

Loads SentenceTransformer model + generates embeddings.

“Turns each PR text into vectors so we can compare similarity.”

Block 9

Custom clustering using cosine similarity + Union-Find at a threshold.

“We cluster PRs based on similarity — if similarity > threshold, put them together.”

This is your **Agglomerative-like custom clustering**.

Block 10

Extracts top keywords from titles in each cluster.

“Shows main words in each cluster so we can understand topic.”

Block 11

Splits PR labels.

“Break labels into a list.”

Block 12

Counts file paths touched per cluster.

“Helps see which folders PRs in each cluster touched.”

Block 13

Streamlit sidebar UI controls.

“User enters repo name, model name, threshold, press buttons.”

Block 14

Streamlit session variables.

“Stores data in app memory so outputs don’t disappear.”

Block 15

Fetches PR data when button pressed.

“Step 1: Download PRs from GitHub.”

Block 16

Embeds PRs when button pressed.

“Step 2: Convert PR text to embeddings.”

Block 17

Clusters and shows cluster insights.

“Step 3: Cluster PRs and show cluster results.”

Block 18

Search bar to find similar PRs.

“You type text and system finds similar past PRs.”

This becomes a **developer assistant search tool**.

Block 19-24

Raw Python (duplicate logic outside Streamlit) to:

- Download PR data
- Save JSON + CSV
- Compute embeddings
- Save embeddings
- Cluster (threshold 0.92)
- Save clustering results

“Same pipeline but run offline for research.”

Block 25-31

Plots & statistics of clusters.

“Visualizes how many clusters, cluster sizes, distribution.”